Sequence-to-Sequence Models and Attention

Scaler School of Technology | Deep Learning I | Week 6 — Session 2/2 | May 2026

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What Is This Lecture About?

Last session you learned LSTMs and GRUs for processing sequences. This session shows how to use them to **transform one sequence into another** — the encoder-decoder framework — and introduces **attention**, the mechanism that directly leads to Transformers next week.

1 Encoder-Decoder & The Bottleneck Problem

Two RNNs connected by a single "context vector" — one reads, one writes. This works for short sequences, but breaks down for long ones because you can't compress an entire sentence into one fixed-size vector. *Like summarizing a book into a single Post-it note.*

2 Attention Mechanism (Bahdanau & Luong)

The fix: let the decoder "look back" at any part of the input at each generation step. It learns to align output words with relevant input words automatically. Bahdanau uses a small neural network for scoring; Luong simplifies it to a dot product. *Like an open-book exam instead of closed-book.*

3 Attention as Soft Lookup — Query, Key, Value

The modern framing of attention: a query (what you want) searches through keys (what's available) and retrieves a weighted mix of values. This is **exactly** the mechanism Transformers use — understanding it here sets you up for next week.

What to Expect in the Session

- Building a seq2seq model for date format conversion
- ⋆ **Visualizing attention heatmaps** — see which input positions the model looks at
- Comparing models with and without attention
- Implementing Bahdanau (additive) attention from scratch
- 2 quizzes on the bottleneck and attention mechanics
- Preview of the Query-Key-Value framework for Transformers

How to Prepare (Optional but Helpful)

- ✓ **Recall** how GRU hidden states work from Session 11
- ✓ **Recall** the softmax function — we'll use it to normalize attention scores
- ✓ **Think about:** "If you were translating a sentence word by word, when would you glance back at the source text?"
- ✓ **Skim** a Google Translate example and notice how word order changes between languages

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Duration: ∼2 hours **Prerequisites:** RNNs, LSTMs/GRUs, hidden states (Sessions 10–11) **Instructor:** Priyansh Saxena